AI for Anomaly Detection

Lecture 1: Introduction to Anomaly Detection



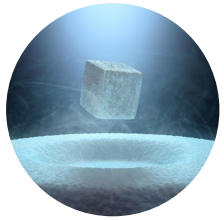


Lilach Goren Huber
Institute for Data Science (IDS)

zh aw School of Engineering

About Me

Dr. Lilach Goren Huber



PhD theoretical physics, Weizmann Institute, Israel

Modelling high temperature superconducting materials.



Physicist + Data Scientist, Pentalum Ltd., Israel

Algorithms for LiDAR-based wind measurements.



Since 2013: Senior Lecturer for Machine Health Intelligence.

- ✓ Machine Learning / Deep Learning
- ✓ Anomaly detection in industrial and infrasturcture systems.
- ✓ Fault diagnostics and predictive maintenance.

Machine Health Intelligence Research @ ZHAW

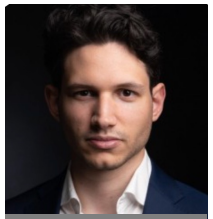
Senior Lecturers


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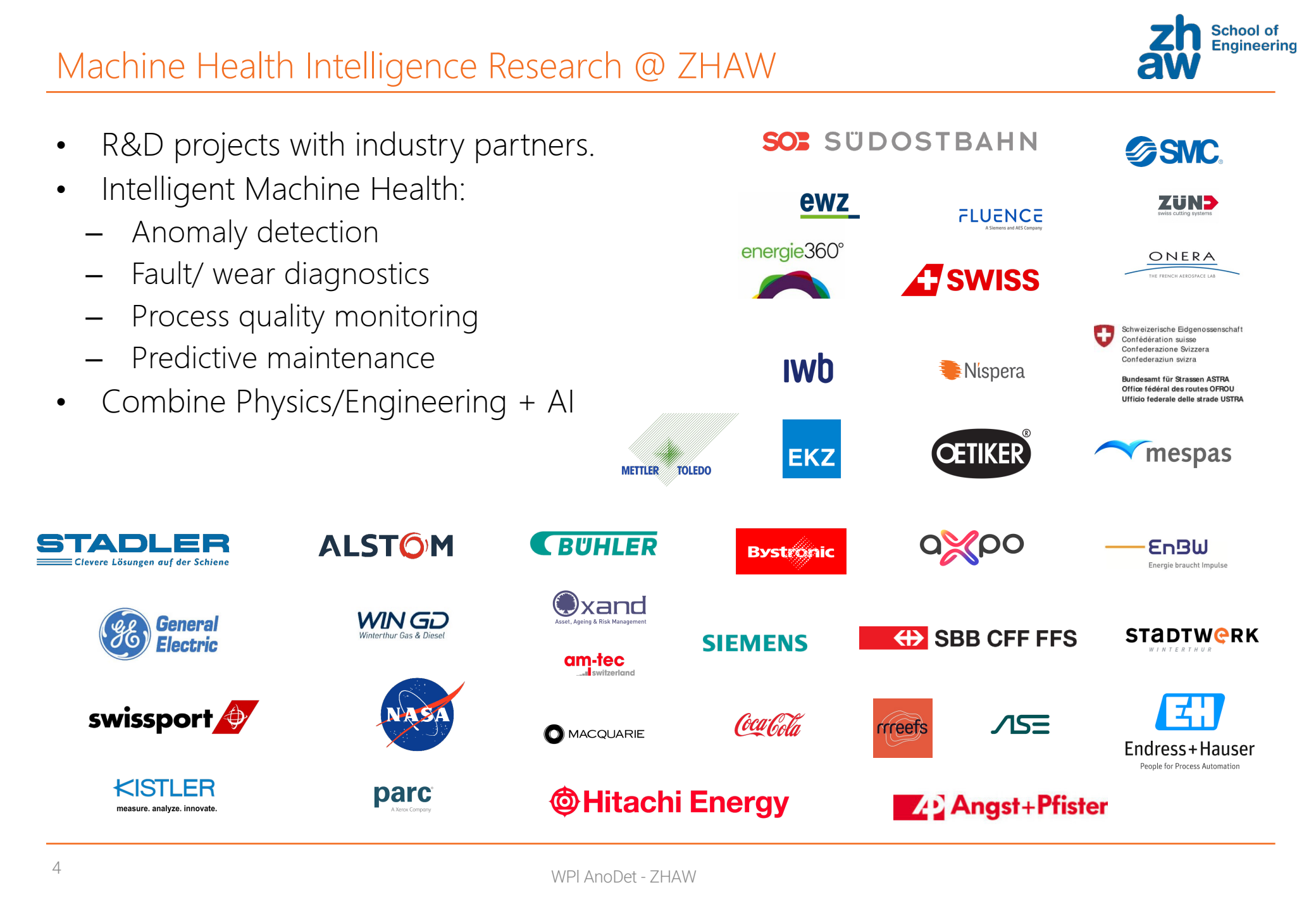
TU Delft TU Delft TU Delft ETH zürich Cranfield University

SWISS parc A Xerox Company Hitachi Energy Endress+Hauser People for Process Automation

MHi MACHINE HEALTH INTELLIGENCE GROUP

Machine Health Intelligence Research @ ZHAW

- R&D projects with industry partners.
- Intelligent Machine Health:
  - Anomaly detection
  - Fault/ wear diagnostics
  - Process quality monitoring
  - Predictive maintenance
- Combine Physics/Engineering + AI



#	Date	Topic
1	15.09	Introducation to anomaly detection (AD)
2	22.09	Supervised ML for AD
3	29.09	Unsupervied ML for AD I
4	06.10	Unsupervised ML for AD II
5	13.10	Residual models: reconstruction ML methods
6	20.10	Residual models: reconstruction neural networks
7	27.10	Residual models: regression models
8	03.11	Initial assignment presentations – obligatory presence in person
9	10.11	Residual models: advanced regression neural networks (CNNs, TCNs)
10	17.11	Domain shifts and data scarcity: transfer learning
11	24.11	Contamination and mislabeling
12	01.12	Quantifying uncertainty, physics informed ML
13	08.12	Recent advances in AD research
14	15.12	Final assignment presentations – obligatory presence in person

Course structure

A typical class contains:

1. Theory
   - learning statistical, machine learning and deep learning techniques for anomaly detection
   - Understanting the math behind, but focussing on intuitive understanding
   - Understanding advantages and limitations of each technique
   - Methodological and scientific approach to problem formulation, solution and evaluation
2. Practical examples
   - Python notebooks demonstrating model development.
3. Exercises
   - Practical supervised work at class (modeling, coding, discussing).

- Questions about your assignments must wait **until the next class**, unless **very** urgent. In urgent cases use my email: gorn@zhaw.ch.

→ the course is designed to be **in-person.**

Course material

On Moodle you find for each class

1. The slides
2. The python notebooks

- Material appears shortly before the class.
- Solution notebooks will be uploaded **after** the class.

Course time

- 16:40-17:25
- 17:30-18:15
- 18:25-19:10 instead of 18:40-19:25
- 19:15-20:00 instead of 19:30-20:15

(Breaks may shift if needed.)

Course grading

During the course you will progressively work on one assignment in groups of 3 students.

The assignment: anomaly detection on a selected dataset.

I. 20% → Initial assignment presentation (3.11).

II. 20% → Final assignment presentation + handing in of presentation and code (15.12).

In each presentation:

- 15-20min for each group.
- Each student is expected to participate **actively,** and answer questions.
- The grade is **individual per student.**

III. 60% → Exam on assignment presentation (January 2027).

- 10-15min for each student
- Each student is expected to know **all aspects of the assignment.**
- There will be questions on the **course content**, beyond your own assignment.
- By construction, each student will face **different questions** ("fairness" cannot be measured).

Details about the projects → next week.

Benefits of this course

- Exposure to realistic use cases and data sets
- Confronting real world challenges of working with data
- Beyond theory: guided hands-on sessions
- Understanding the added value of a data scientist on top of LLMs and agentic coding.

Who should NOT attend this course?

- Students wishing rigorous math proofs
- Students wishing no mathemtical / technical depth at all
- Students expecting only domain-specific examples
- Students wishing a minimal effort ☺

What do you need to start?

- Basic knowledge of statistics and probability:
  - Distributions, correlations.
  - Mean, std, median, quantiles.
  - Linear regression.





- Basic knowledge of python:
  - Numpy
  - Matplotlib
  - Pandas
  - Sklearn ➔ will further develop in this course.

- A google account to be able to work on colab.



Anomaly Detection (AD)

"The identification of rare items, events or observations which deviate significantly from the majority of the data and do not conform to a well-defined notion of normal behaviour."

Why are we interested in AD?

1. Anomalies can be **interesting** (faults, attacks, diseases) ➔ further investigate them.



2. Anomalies can **bias** models (noise, errors, mislabelling)➔ remove them.



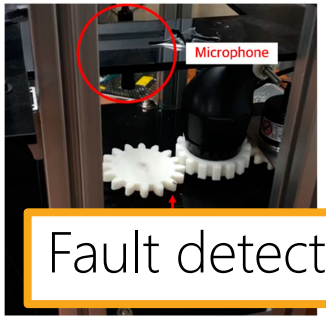
Where do we encounter anomaly detection?

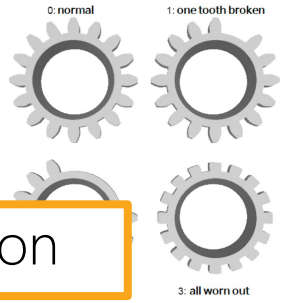
Do you have any examples?

Why does anomaly detection matter?



Cyber-attack detection





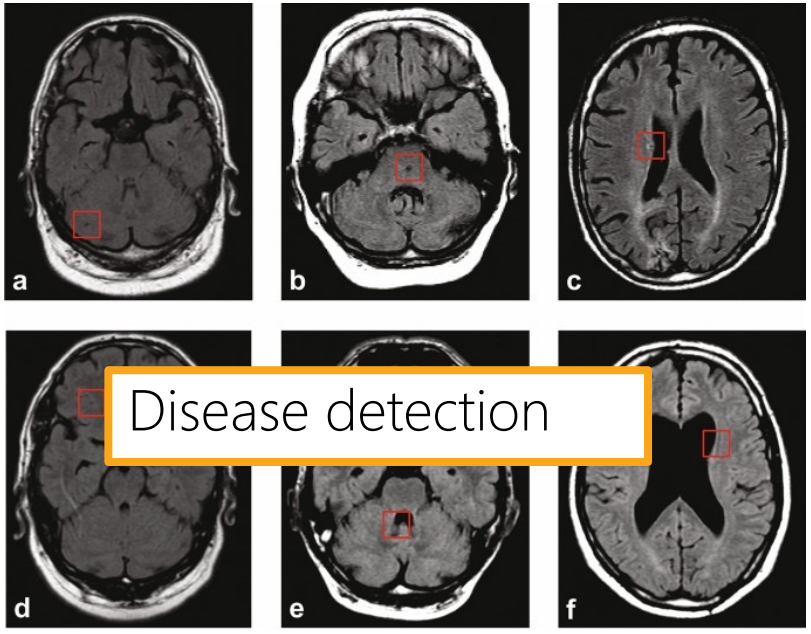
Fault detection



Fraud detection

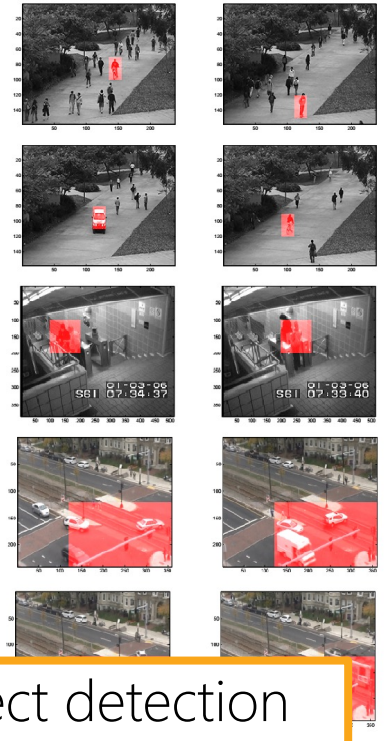


Abnormal behaviour detection



Disease detection





Object detection

Kees M. van Hespen et al, An anomaly detection approach to identify chronic brain infarcts on MRI (2021).
Jonghoek Kim et al.,A Study on Gear Defect Detection via Frequency Analysis Based on DNN (2022).
https://towardsdatascience.com/credit-card-fraud-detection-9bc8db79b956.
Saligrama & Chen, Video Anomaly Detection Based on Local Statistical Aggregates, CVPR (2012).
https://www.passwordboss.com/what-to-do-if-you-experience-a-data-breach/
https://alwaysai.co/blog/anomaly-detection-manufacturing

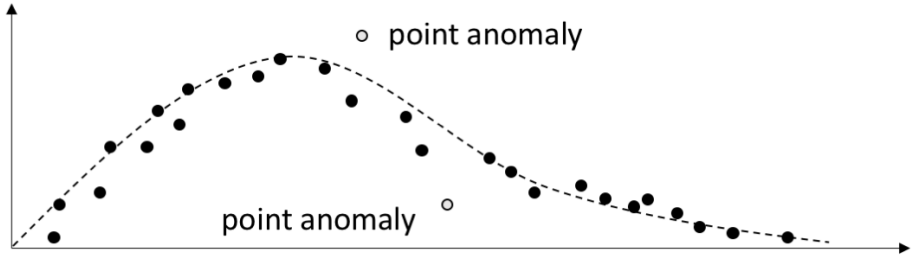
What is an anomaly?

A data point that deviates significantly from the norm.

Types of anomalies:

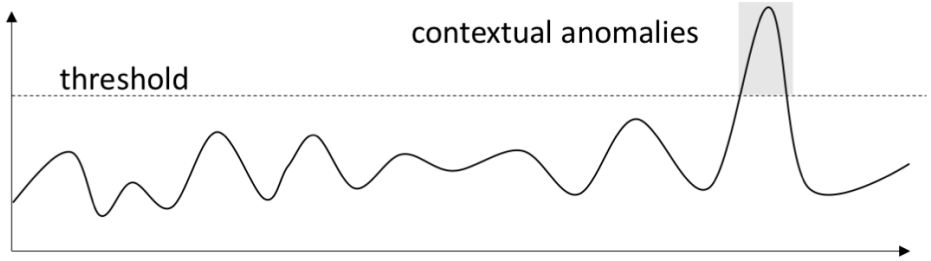
- **Point anomalies (outliers):**

A single data point that deviates significantly from the rest of the data.



- **Contextual anomalies (novelties):**

A data point that is normal in one context but anomalous in another.



- **Collective anomalies (novelties):**

A group of data points that collectively deviate from the norm, though individual points may not be anomalous on their own.

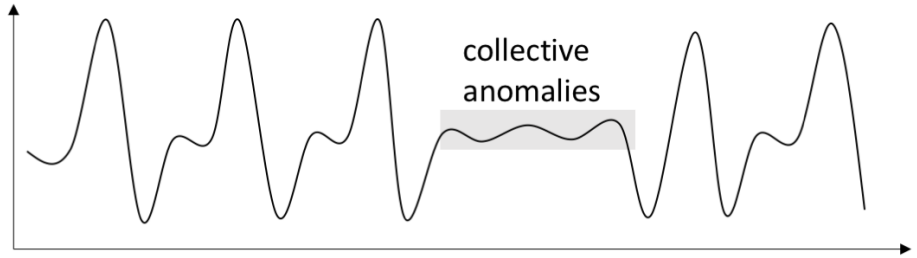
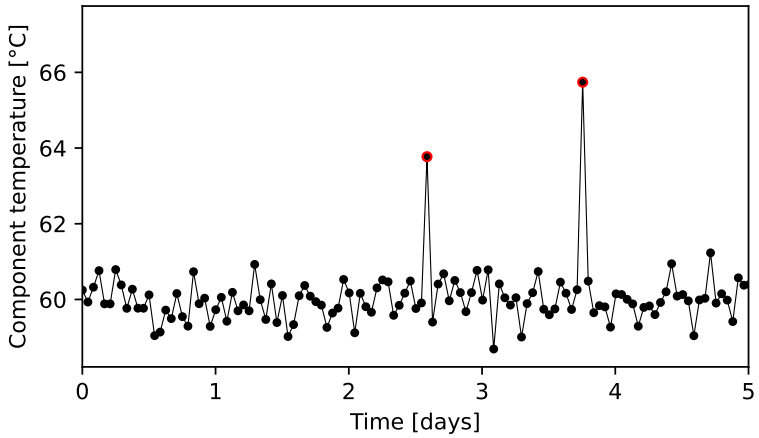


Image source: https://www.oreilly.com/library/view/hands-on-industrial-internet

Point anomalies (outliers)

A single data point that deviates significantly from the rest of the data.

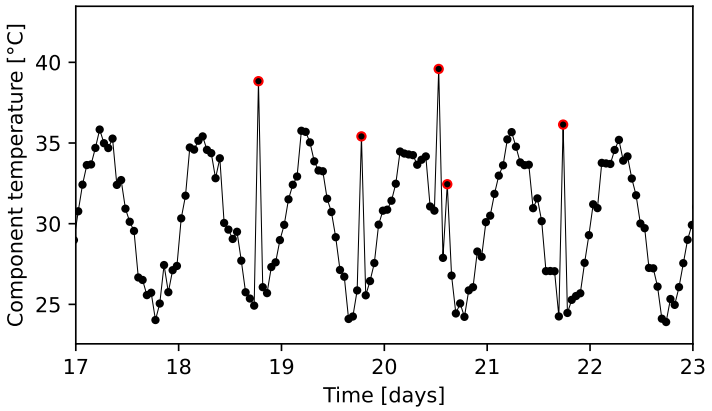
- Finance: transaction which is orders of magnitude higher.
- Healthcare: one patient with a very high blood pressure.
- Manufacturing: temperature sensor reading of 300°C in a machine with a typical operating temperature of 50-100°C.



Contextual anomalies (novelties)

A data point that is normal in one context but anomalous in another.

- Weather data: a temperature of 35°C is normal in summer but anomalous in winter.
- E-Commerce: high shopping activity late at night is normal on Black Friday but anomalous otherwise.
- Healthcare: heartbeat of 120 is normal during sports but anomalous at rest.



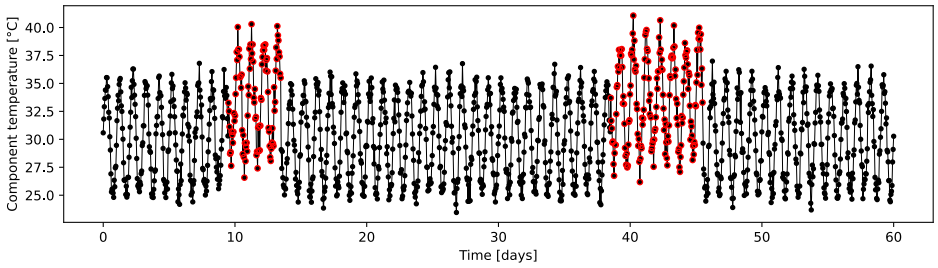
Collective anomalies

A group of data points that collectively deviate from the norm, though individual points may not be anomalous on their own.

- Sensor data in manufacturing: A batch of images shows slight deformities in each product. e.g., misaligned labels or subtle cracks. While no single image is severely anomalous, the collection reveals a systematic production issue.

- Fraud detection: Multiple small purchases on a credit card within a short time frame, which could indicate fraud.

- Medical: Heart skips a beat.



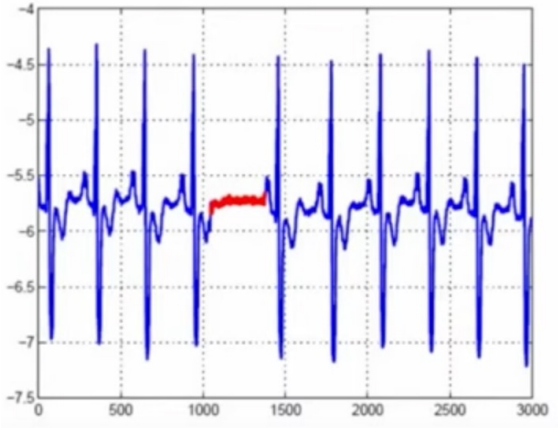


Image source: A review of ML techniques for AD, David Green

Challenges in anomaly detection

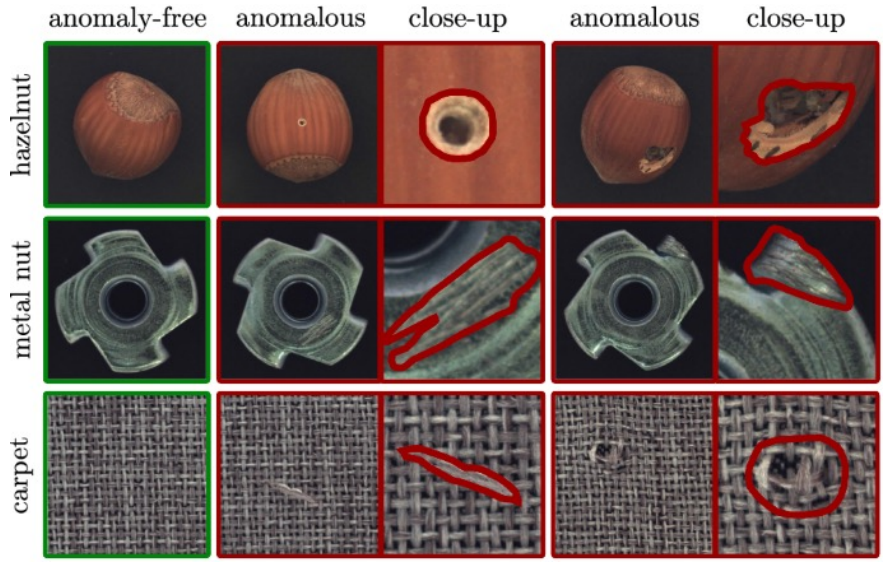
- Anomalies are rare. Often very few or no past examples ➔ imbalanced data.

- Anomalies are heterogeneous. New kinds of anomalies can show up, with different characteristics and time scales ➔ detect known and **unknown** anomaly types.

- Anomalies are rarely labelled. In complex systems, historical data is often partial, not understood, information no longer available ➔ **validation** and **evaluation** is inherently complicated.

➔Anomaly detection is **fundamentally different from classification.**

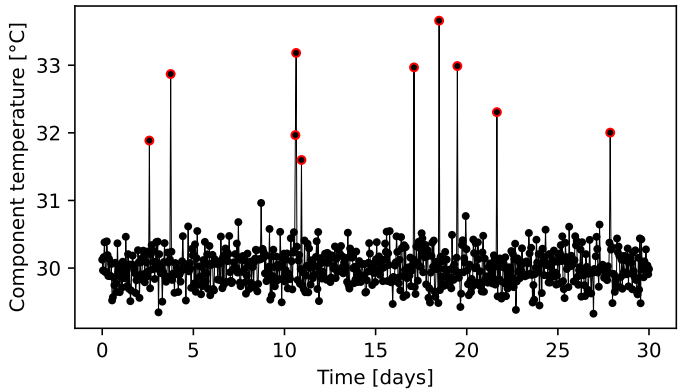
➔ requires its own strategies.



Source: MVTEC AD dataset, Bergmann et al. 2021.

Systematic Anomaly Detection

- Example: temperature of an isolated system.
- A simple threshold would do.
- How to decide on a threshold?



Define an outlier/anomaly:

"an observation which deviates so much from other observations as to arouse suspicions that it was generated by a different mechanism"

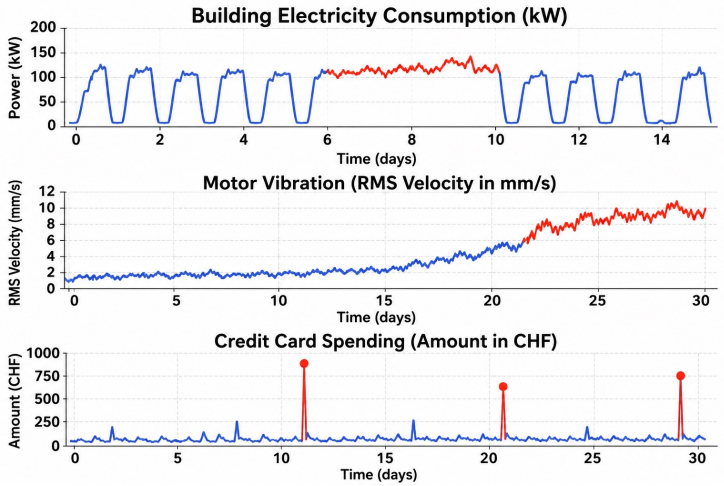
(D. Hawkins, 1980).

➔Model the regular mechanism, e.g as a Gaussian distribution.

➔If a point is at the extreme tail, it is likelier to be generated by another distribution.

AD in univariate data

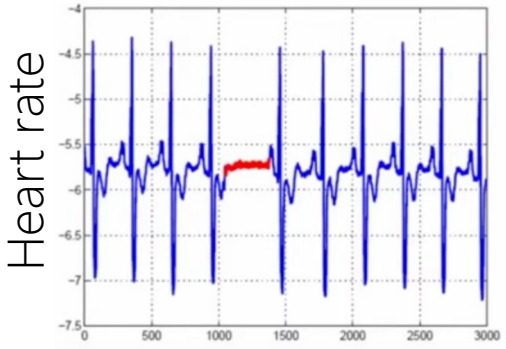
- Building electricity consumption
  - Normal: clear day/night pattern.
  - Anomaly: air conditioning left on over weekend.



- Motor vibration
  - Normal: nearly constant level.
  - Anomaly: gradual increase due to bearing wear.

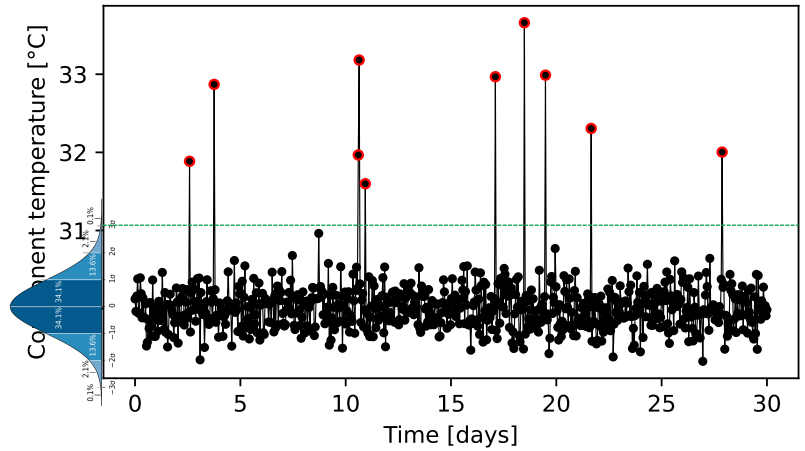
- Credit card transaction amount
  - Normal: many small purchases.
  - Anomaly: one unusually large transaction (possible fraud).

- Heart rate
  - Normal: regular pattern.
  - Anomaly: heart skips a beat.

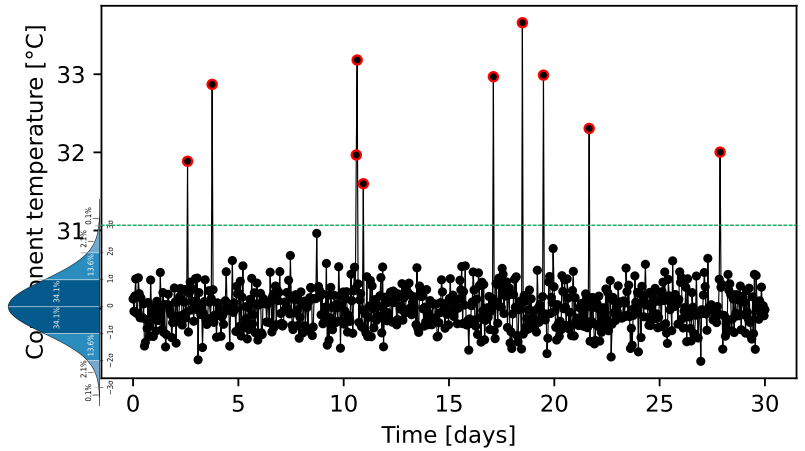


AD in univariate data

- Example: temperature of an isolated system.
- How to decide on a threshold?
- Use the distribution of “regular” temperatures



- Example: temperature of an isolated system.
- How to decide on a threshold?
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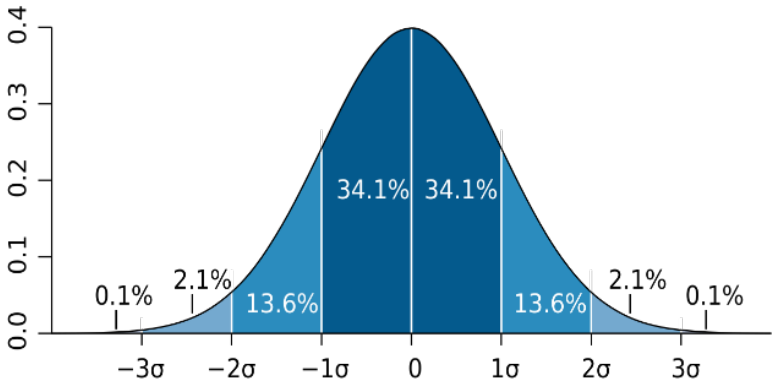


- Model the temperatures with a Gaussian distribution.

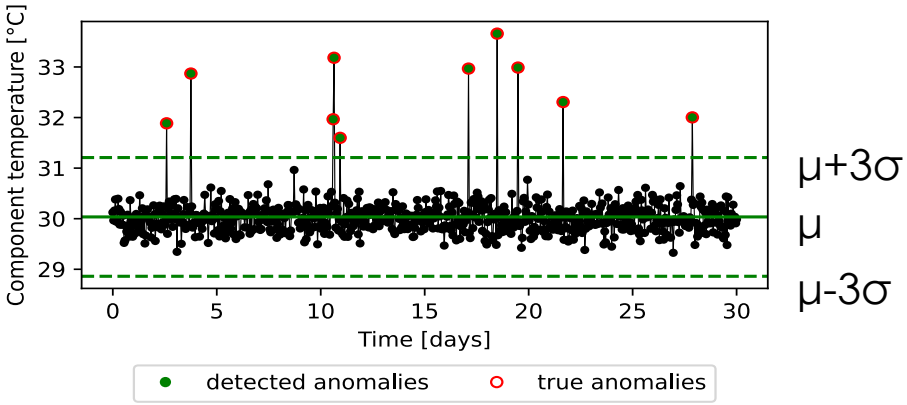
➔ 68% of the values are in (μ-σ, μ+σ)

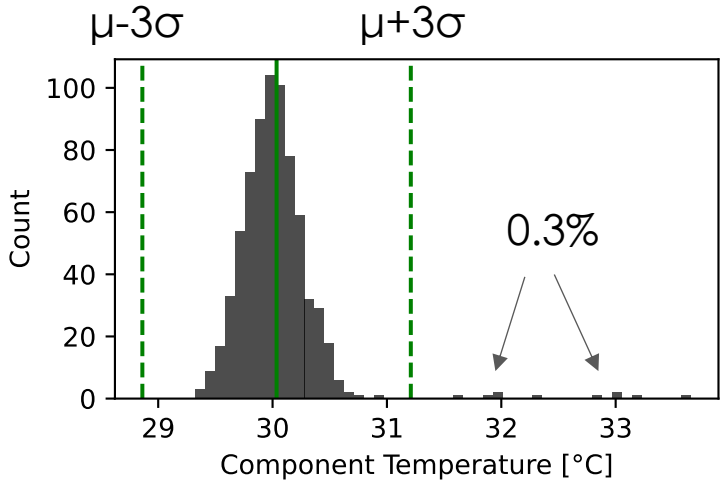
95% are in (μ-2σ, μ+2σ)

99.7% are in (μ-3σ, μ+3σ)



Statistical threshold





- Model the temperatures with a Gaussian distribution.

➔ 68% of the values are in (μ-σ, μ+σ)

95% are in (μ-2σ, μ+2σ)

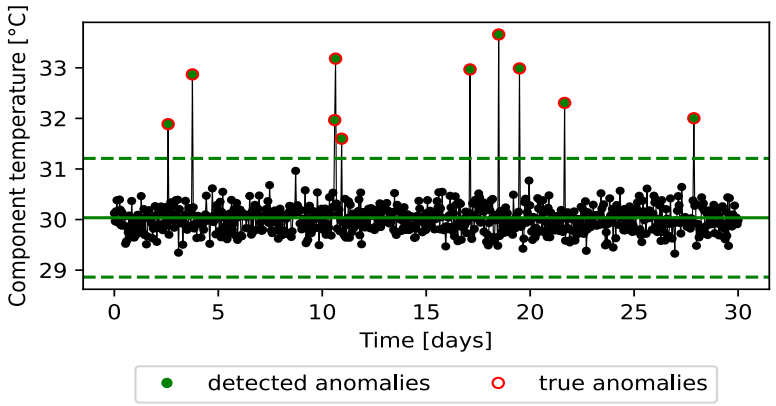
99.7% are in (μ-3σ, μ+3σ)

Statistical threshold

- Assume that normal operation follows a Gaussian distribution.

$$H_0: \; x \sim \mathcal{N}(\mu, \sigma^2)$$

- Compute the tail probability of a new observation.

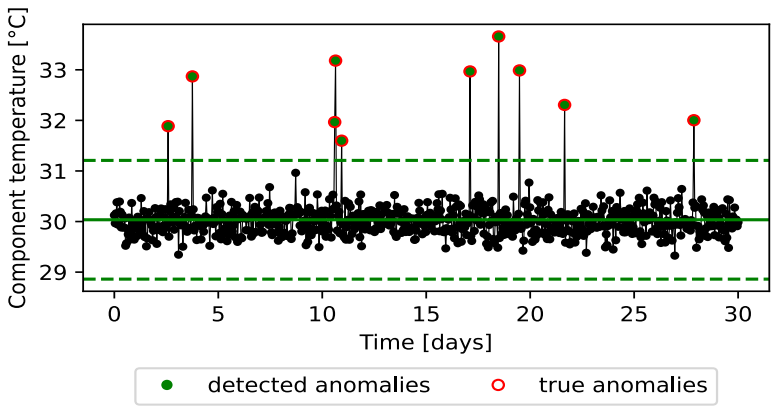


Statistical threshold

- Assume that normal operation follows a Gaussian distribution.

$$H_0: \; x \sim \mathcal{N}(\mu, \sigma^2)$$

- Compute the tail probability of a new observation.



One sided detector

$$p_i = 1 - \mathrm{F}(x_i; \mu, \sigma)$$

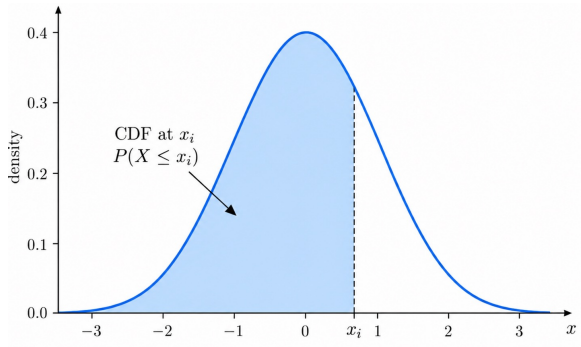
Two sided detector

$$p_i = 2\min\{\mathrm{F}(x_i), 1 - \mathrm{F}(x_i)\}$$

Application relevant anomalies!

Cumulative Distribution Function (CDF)

Large p-value: observation is consistent with normal behavior.

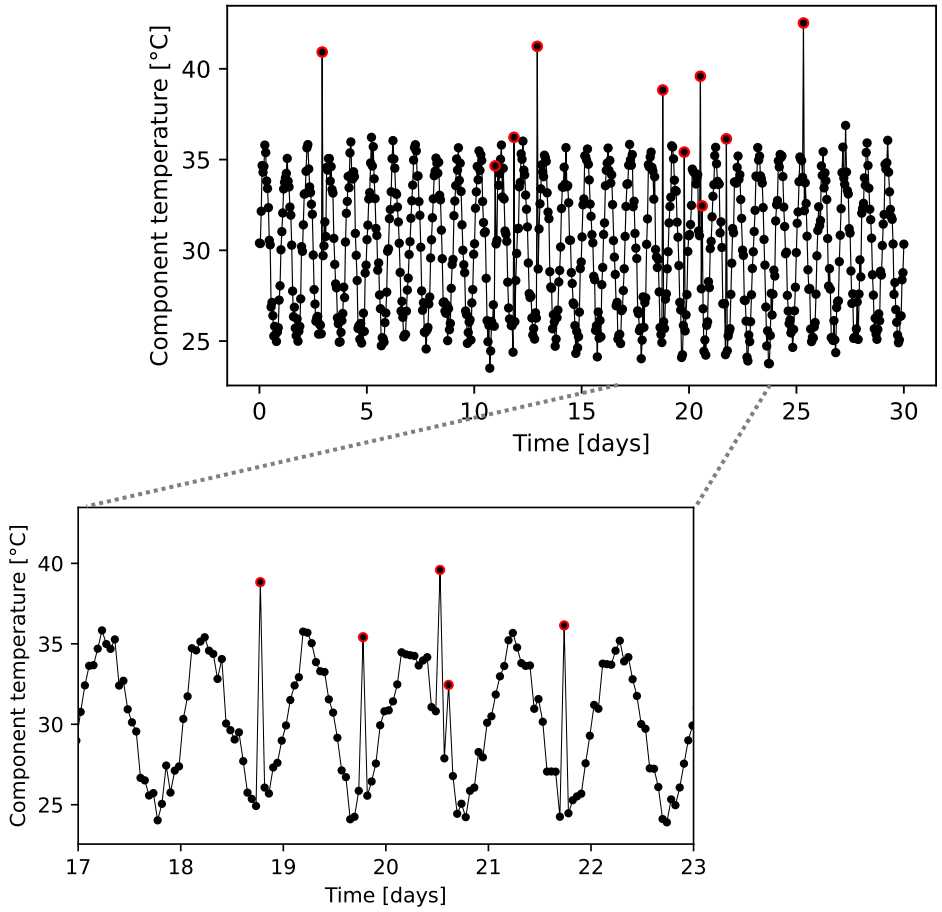


Small p-value: observation is unusual under the normal model.

If $\mathbf{p < \alpha}$, flag the observation as an anomaly.

Example: component temperature of a wind turbine.





Goal: early detection of high risk failures.

Image source: https://www.norfolklive.co.uk/news/norfolk-news/gallery/video-show-wind-turbine-great-8678852

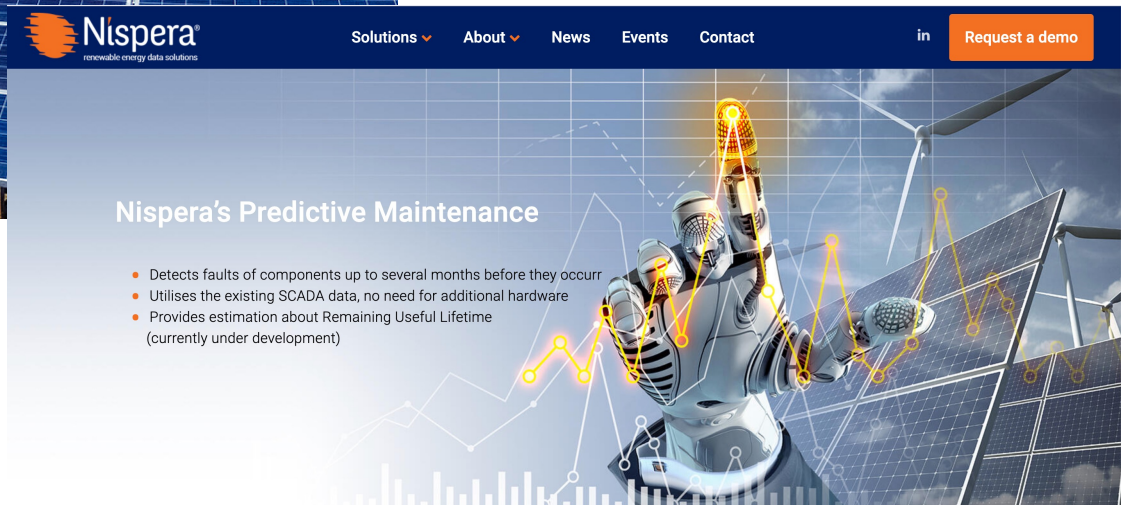
Example: our project with Nispera



Schweizerische Eidgenossenschaft
Confédération suisse
Confederazione Svizzera
Confederaziun svizra

Swiss Confederation

Innosuisse – Swiss Innovation Agency



FLUENCE
A Siemens and AES Company

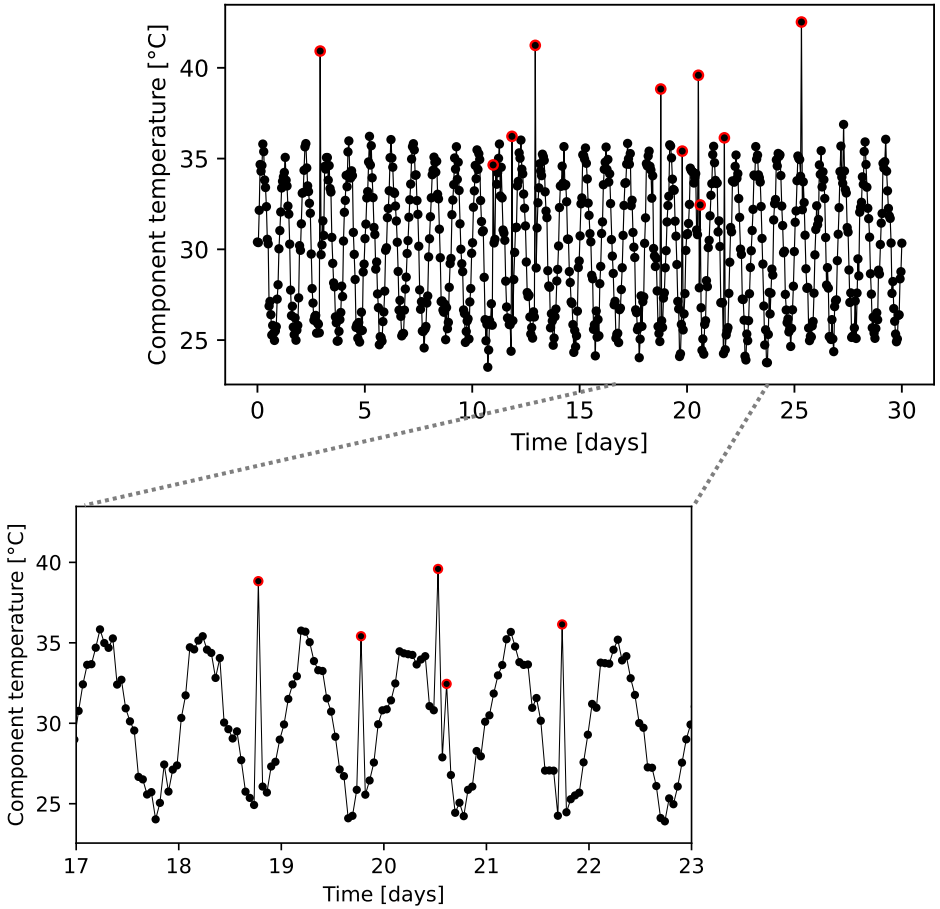
EKZ

Example: component temperature of a wind turbine.



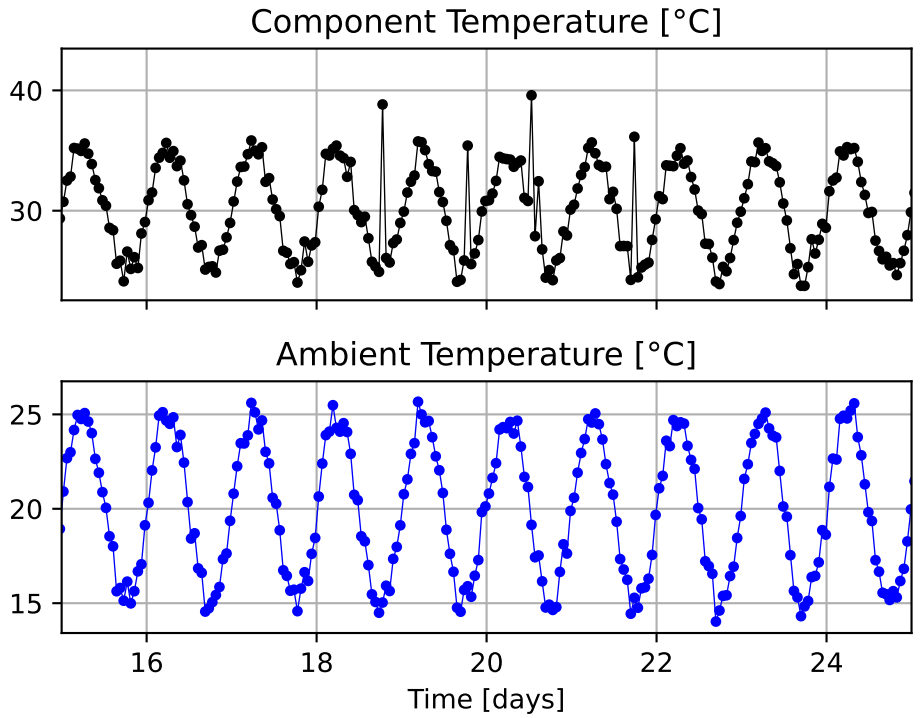
Challenge: simple threshold would not work.

Why?

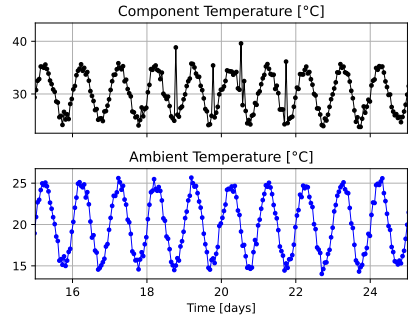


Step 1: visualize all variables.

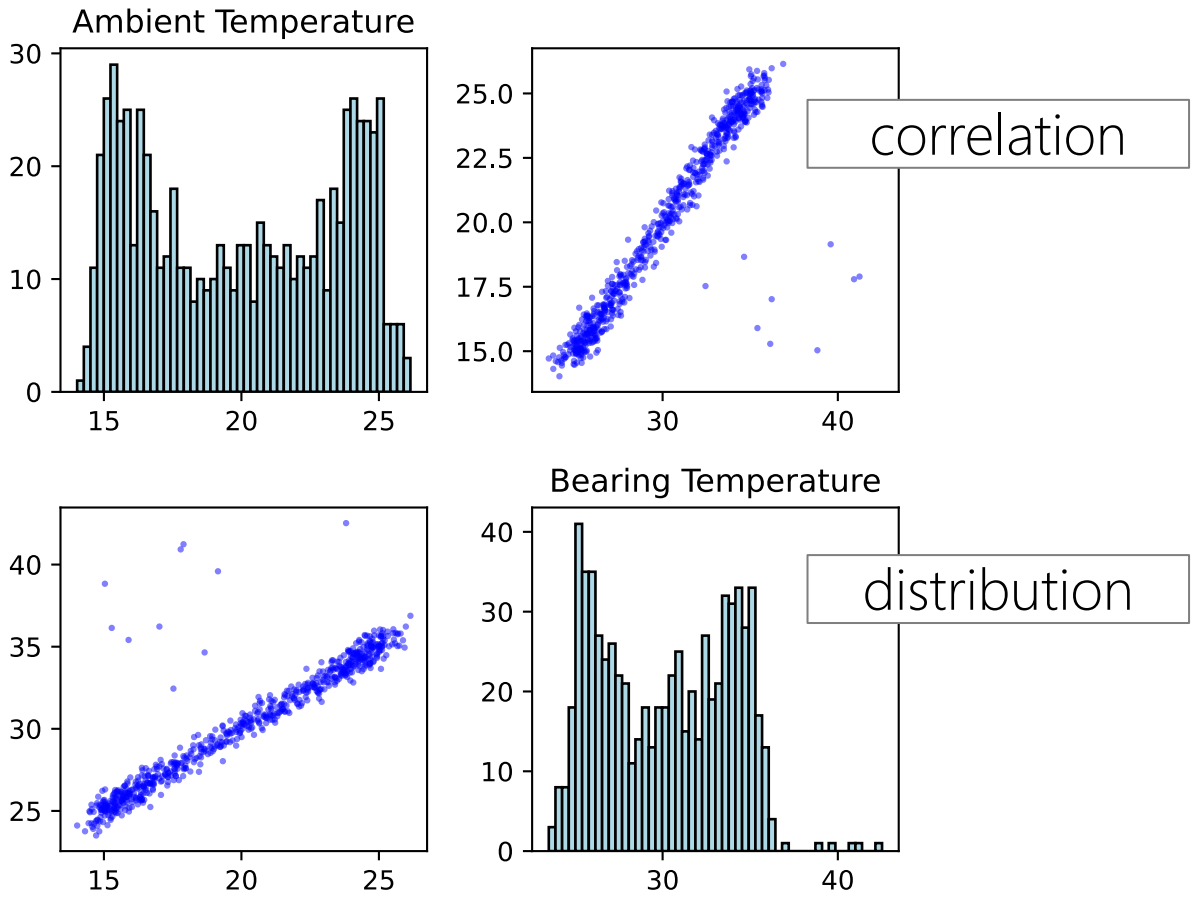
- Understand the “regular” mechanism.
- Visualize all variables that can influence the target variable.
- Use domain knowledge if possible.



Step 2: visualize correlations.



Pair plot

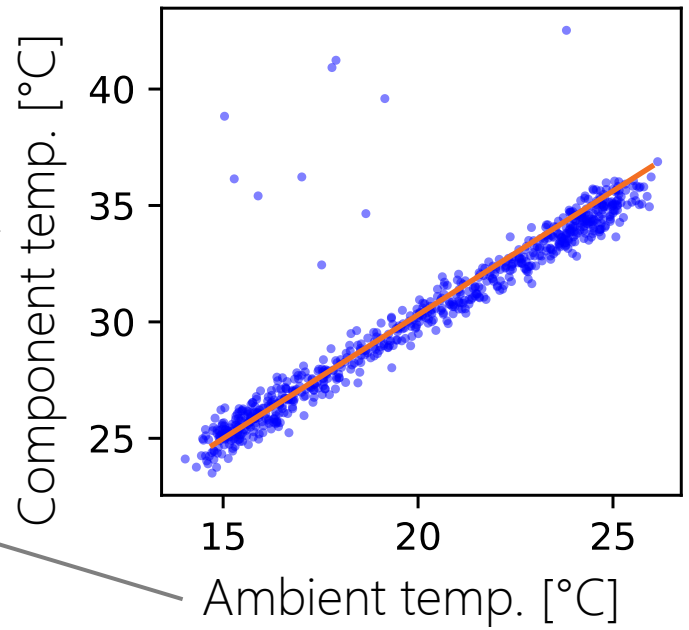


Anomalies do not obey the rules!
Find the rule and look for points that do not obey it.

In this case: linear relationship.
➔Simplest model: linear regression.

$$y = \beta_0 + \beta_1 x$$

Find e.g by least squares



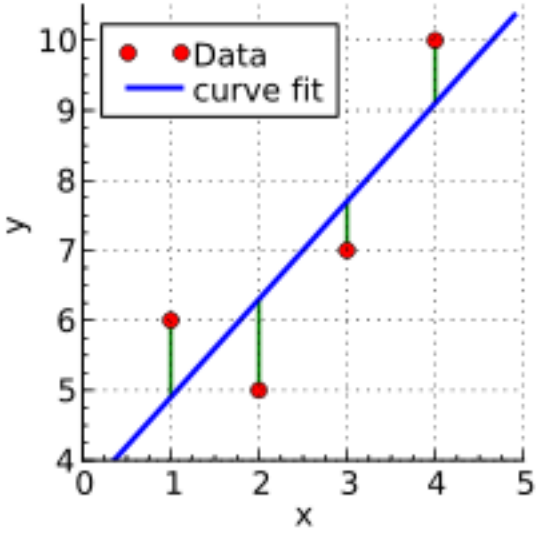
Linear regression recap (Ordinary Least Squares)

- Each point k consists of a set of m input variables $x_{1k} \dots x_{mk}$
- and a continuous outcome variable y_k
- use the dataset to create a predictive model of the outcome variable

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_m x_m$$

- The fit parameters are found by minimizing the Mean Squared Error loss:

$$\mathcal{L}_{MSE} = \frac{1}{N} \sum_{k=1}^{N} |y_k - \hat{y}_k|^2$$



$$y_k = \beta_0 + \beta_1 x_{1k} + \beta_2 x_{2k} + \cdots + \beta_m x_{mk}$$

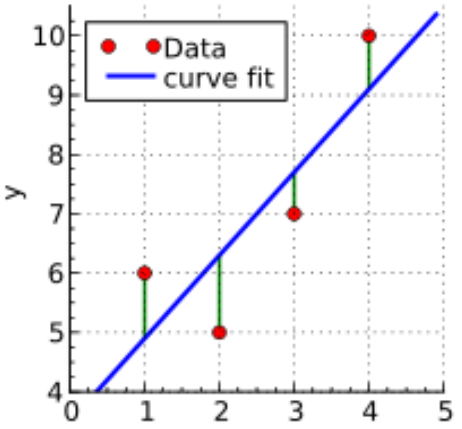
Step 4: Residuals can be used for anomaly detection

We minimized $\mathcal{L}_{MSE} = \frac{1}{N}\sum_{k=1}^{N} |y_k - \hat{y}_k|^2$

Define the residual of measurement k $\varepsilon_k = y_k - \hat{y}_k$

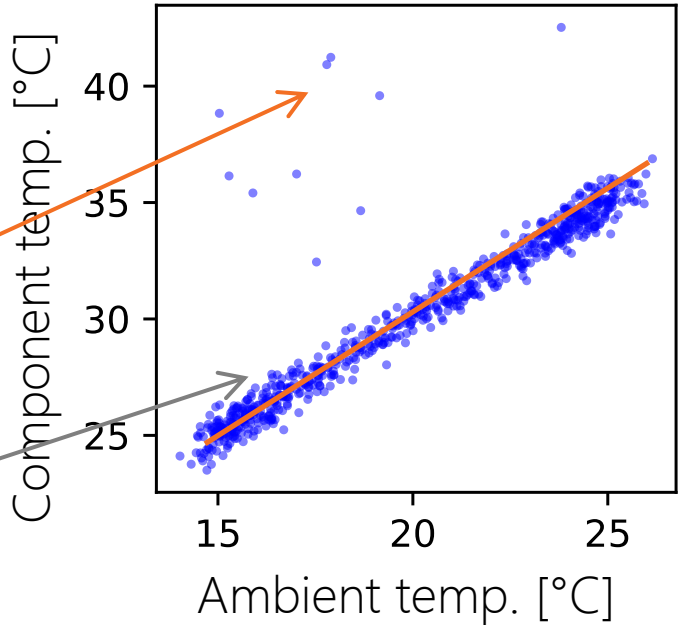
observed

Predicted (fitted)



High residuals

Low residuals



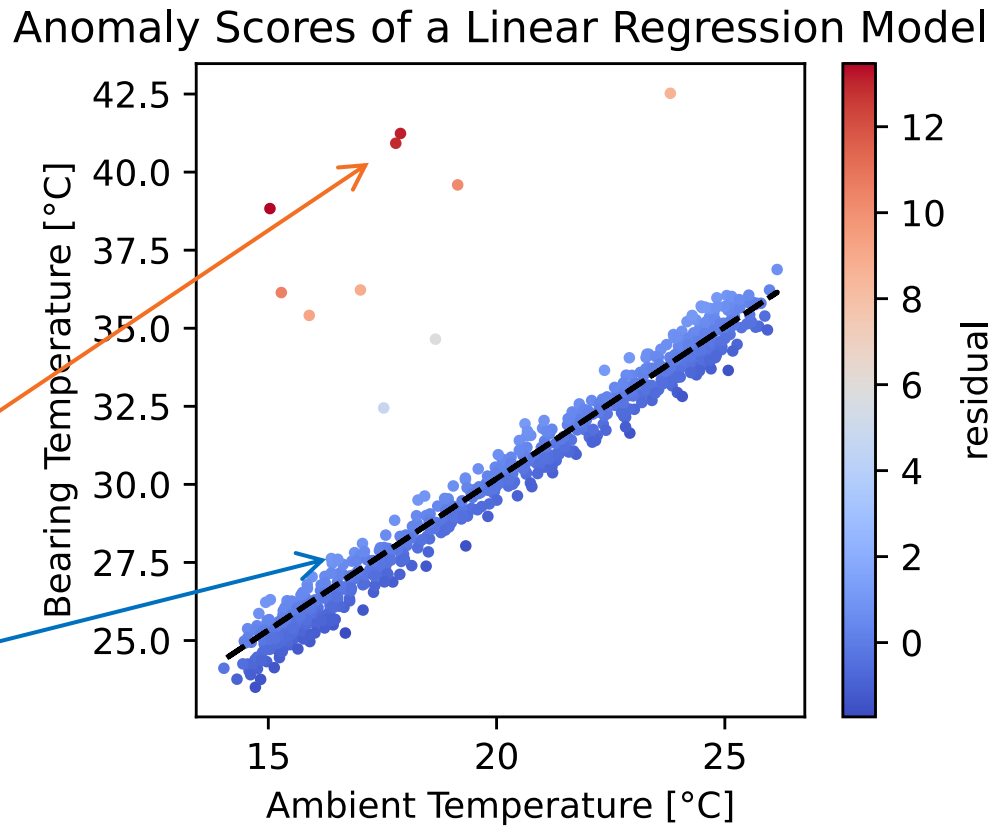
WPI_AnoDet_01b_AD_statistical_solution

Step 4: residuals as anomaly scores

- Observations.
- Model normality.
- Measure deviation from normality = residuals.
- Assign a score to each observation.

High anomaly score

Low anomaly score

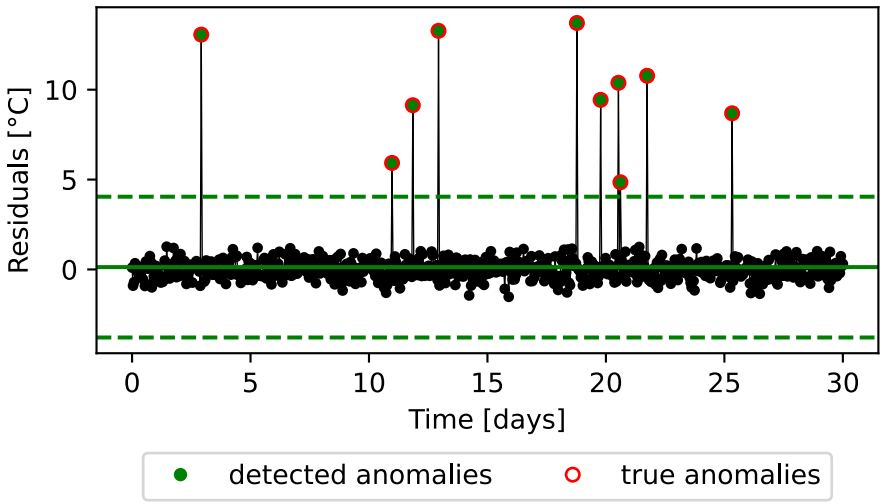


residual of measurement k $\varepsilon_k = y_k - \hat{y}_k$

observed Predicted (fitted)

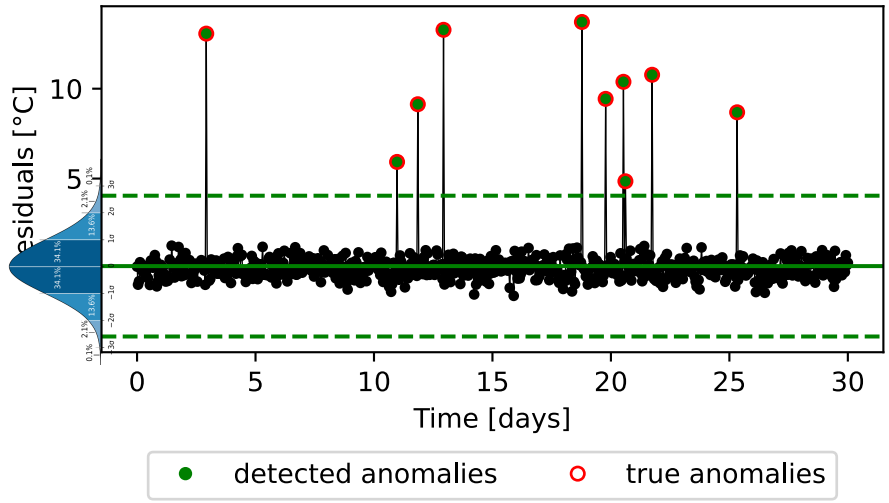
Step 5: select a detection threshold

- Notice that we effectively reduced the 2 dimensional problem to one dimension = the residuals.
- Now we can treat the problem as univariate ➔ same as before.

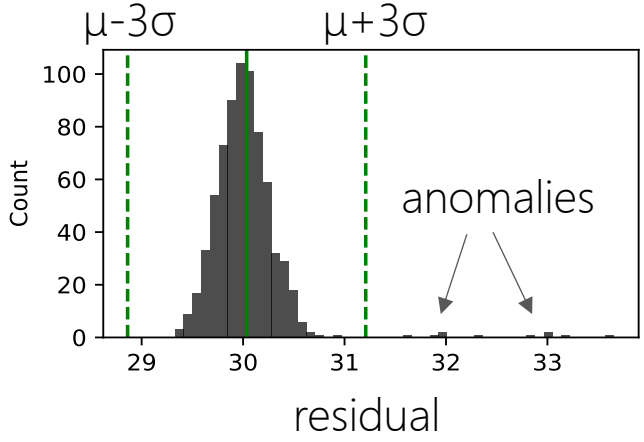


Step 5: select a detection threshold

- Notice that we effectively reduced the 2 dimensional problem to one dimension = the residuals.
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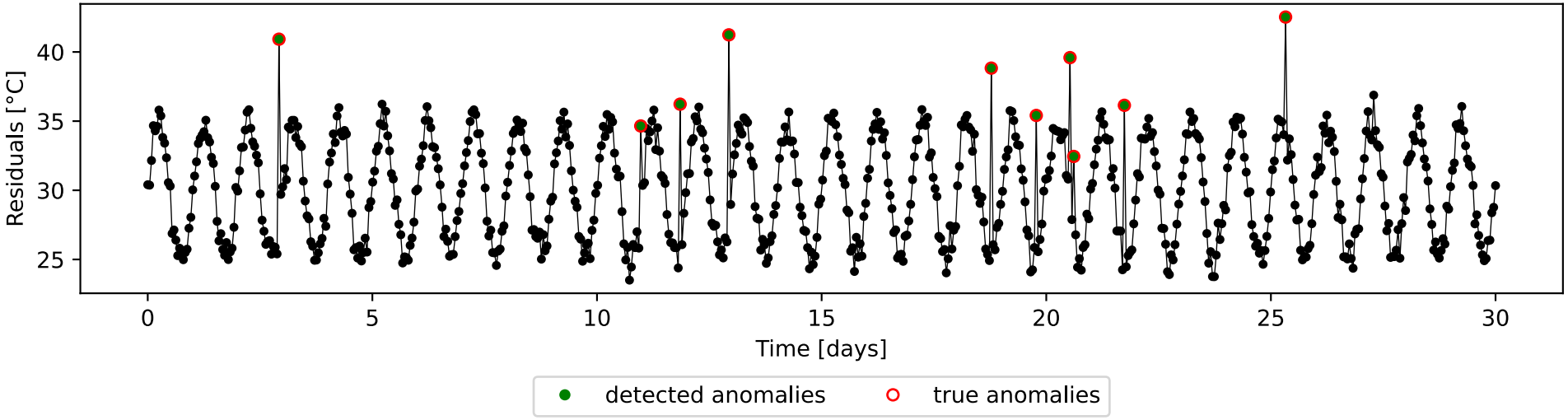


- Set a threshold at the tail of the distribution.



Step 6: sanity check

- Always return to the “real world”.



Step 7: evaluation

- Possible only if we have labels.
- To validate your model you need some labels.

- Note:
  - We did not use the labels to detect anomalies
  - We will now use them to evaluate how well we did.

Evaluation

- Anomaly detection is different from classification.
- Classification:
  - Train a model to assign labels y to samples x. e.g: spam vs. legitimate mail.
  - During training the model recieves both the samples
  - To evaluate/ test the performance: compare predicted labels $\hat{\mathbf{y}}$ to true labels $\mathbf{y}$.

- Anomaly detection
  - Train a model to identify abnormal samples x.
  - During training the model does not always recieve normal/abnormal labels.
  - To evaluate/ test the performance: compare predicted labels to true labels.

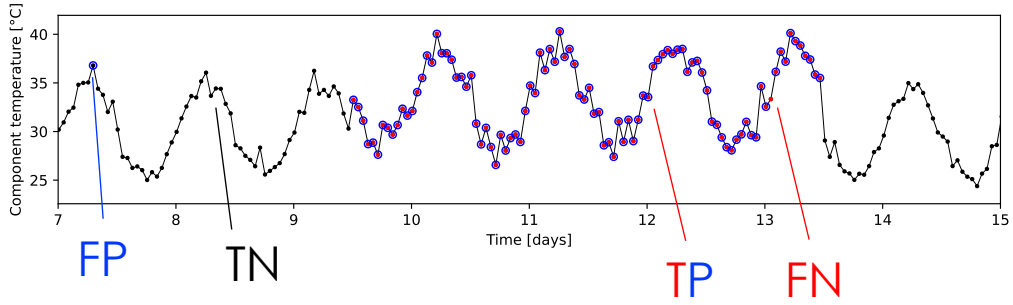
➔ Evaluation is the same as in a binary classification.

		PREDICTED: Positive	PREDICTED: Negative
ACTUAL	Positive	True Positive (TP)	False Negative (FN)
ACTUAL	Negative	False Positive (FP)	True Negative (TN)

Evaluation

- Anomaly detector predictions:
  - Positive: detector predicts “anomaly”
  - Negative: detector predict “normal”

  (Think of Covid…)

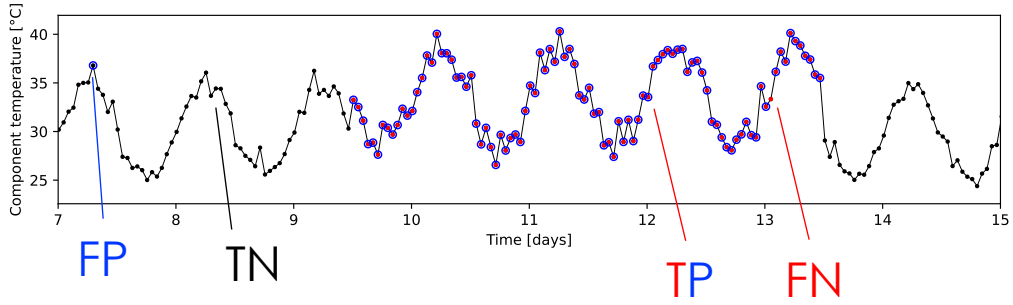


- Compare to true label:
  - TP **True** positive: detector detects **correctly** an anomaly.
  - TN **True** Negative: detector detects **correctly** a normal sample.
  - FP **False** positive: detector **mis**classifies a true normal sample.
  - FN **False** negative: detector **mis**calssifies a true anomaly.

Evaluation

- Anomaly detector predictions:
  - Positive: detector predicts “anomaly”
  - Negative: detector predict “normal”

  (Think of Covid…)



- Compare to true label:
  - TP **True** positive: detector detects **correctly** an anomaly.
  - TN **True** Negative: detector detects **correctly** a normal sample.
  - FP **False** positive: detector **mis**classifies a true normal sample.
  - FN **False** negative: detector **mis**calssifies a true anomaly.

➔ Confusion matrix:

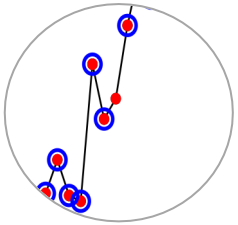
		PREDICTED: Positive	PREDICTED: Negative
ACTUAL	Positive	True Positive (TP)	False Negative (FN)
	Negative	False Positive (FP)	True Negative (TN)

- Tradeoff between 2 types of errors.
  - Low detection threshold ➔ many false alarms (“false positives”)
  - High detection threshold➔ many missed detections (“false negatives”)

What is relevant to ask:

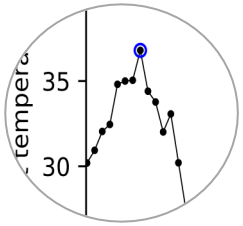
- If there is an anomaly, how probable I am to detect it?

$$Recall = TPR = sensitivity = \frac{TP}{TP + FN}$$



- If there is no anomaly, what is the probability that I classify it wrongly?

$$FPR = 1 - specificity = \frac{FP}{TN + FP}$$



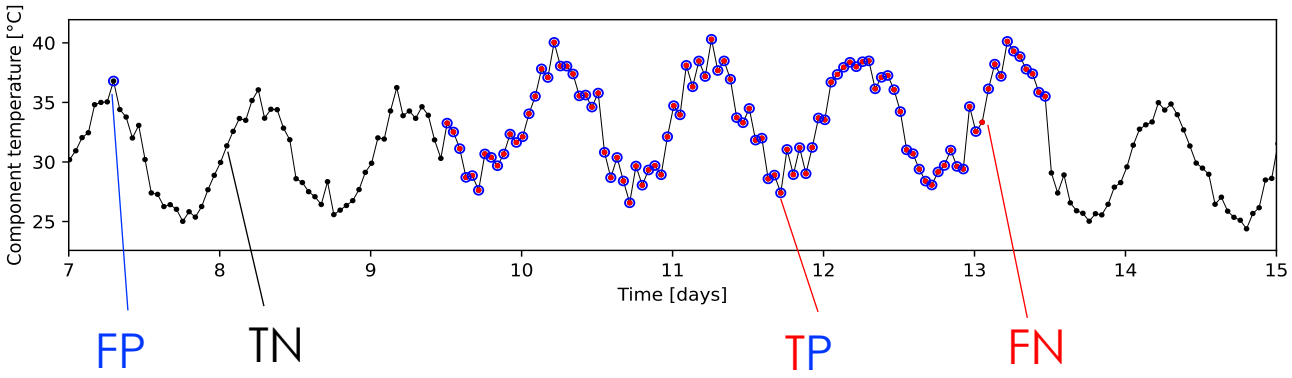
- If I detected an anomaly, what is the probability that it is indeed an anomaly?

$$Precision = \frac{TP}{TP + FP}$$

So it depends on the threshold…?

All of the above evaluation metrics require

- An anomaly detector
- → anomaly score for each sample (e.g regression residuals).
- A decision threshold: which scores are considered anomalous?



Selecting the correct threshold is a challenge in ist own

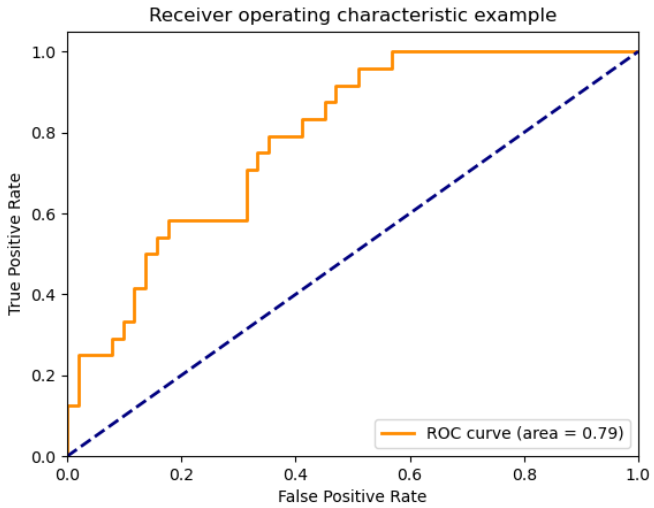
➔can we evaluate the detection performance independently from a threshold?

Evaluation metric that is independent on the threshold

Receiver Operating Characteristic (the name comes from radar signal detection during WWII)

- **Receiver** = the radar receiver.
- **Operating** = the choice of detection threshold.
- **Characteristic** = the resulting performance (detection rate vs. false alarm rate).

$$Recall = TPR = \frac{TP}{TP + FN}$$

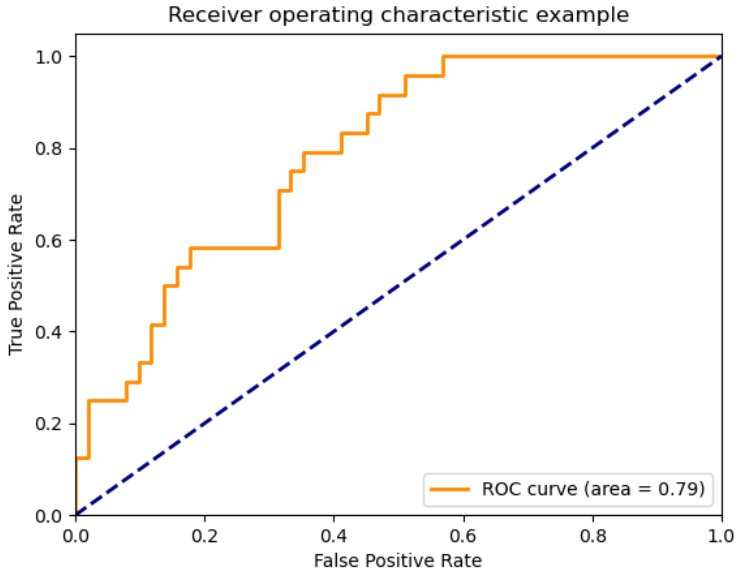


$$FPR = \frac{FP}{TN + FP}$$

ROC curve

1. Compute an anomaly score for every test sample.
2. Sweep the decision threshold from high to low.
3. At each threshold:
   - classify samples as normal/anomaly
   - compute TPR and FPR
4. Plot TPR versus FPR.

$$Recall = TPR = \frac{TP}{TP + FN}$$



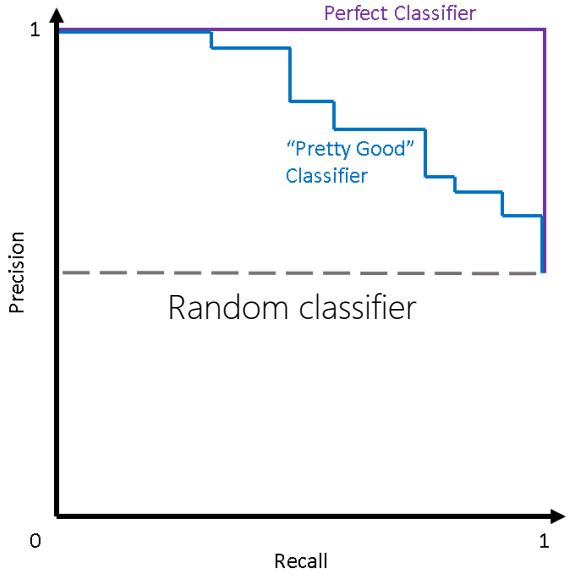
5. Compute the area under the curve (AUROC).

$$FPR = \frac{FP}{TN + FP}$$

Precision-Recall curve (PRC)

1. Compute an anomaly score for every test sample.
2. Sweep the decision threshold from high to low.
3. At each threshold:
   - classify samples as normal/anomaly
   - compute Precision and Recall
4. Plot Precision versus Recall.

$$Precision = \frac{TP}{TP + FP}$$

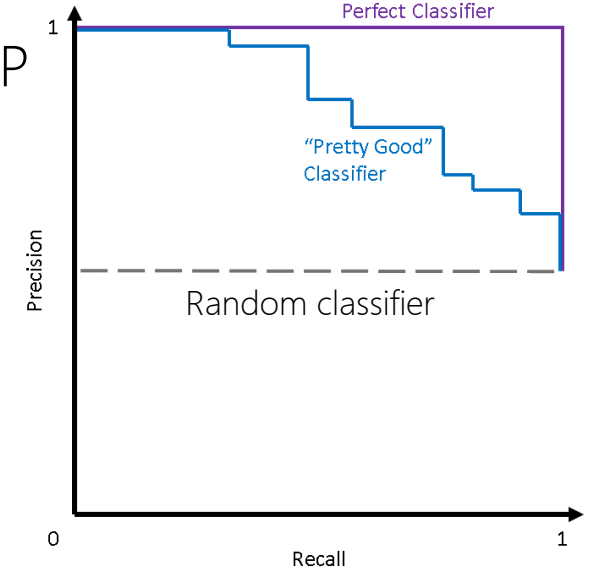


$$Recall = TPR = \frac{TP}{TP + FN}$$

5. Compute the area under the curve (AUPRC=AP).

Threshold independent evaluation metrics

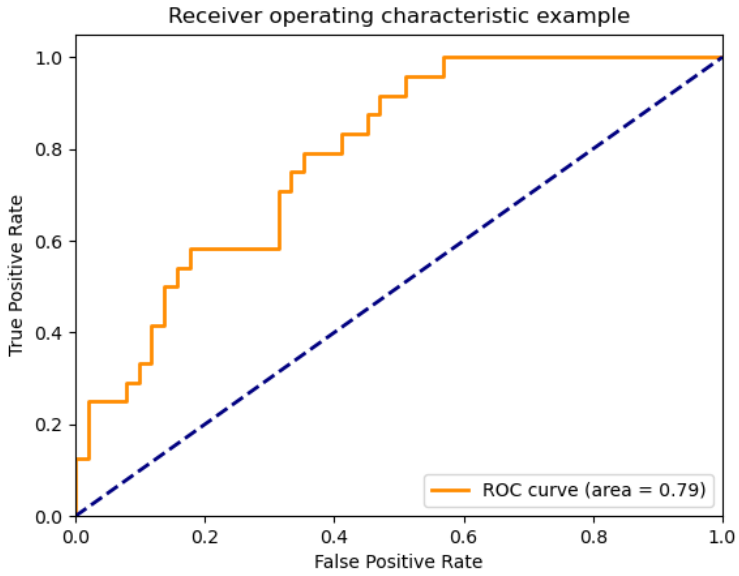
PRC plot➔ AUPRC=AP



$$Precision = \frac{TP}{TP + FP}$$

$$Recall = TPR = \frac{TP}{TP + FN}$$

ROC plot➔ AUROC

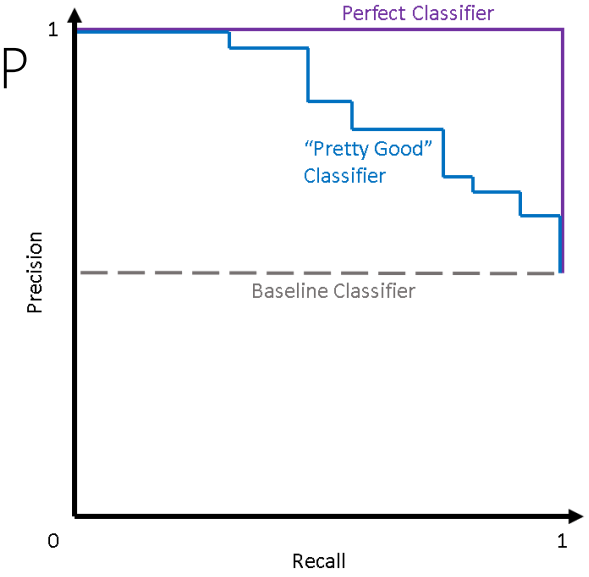


$$FPR = \frac{FP}{TN + FP}$$

Threshold independent evaluation metrics

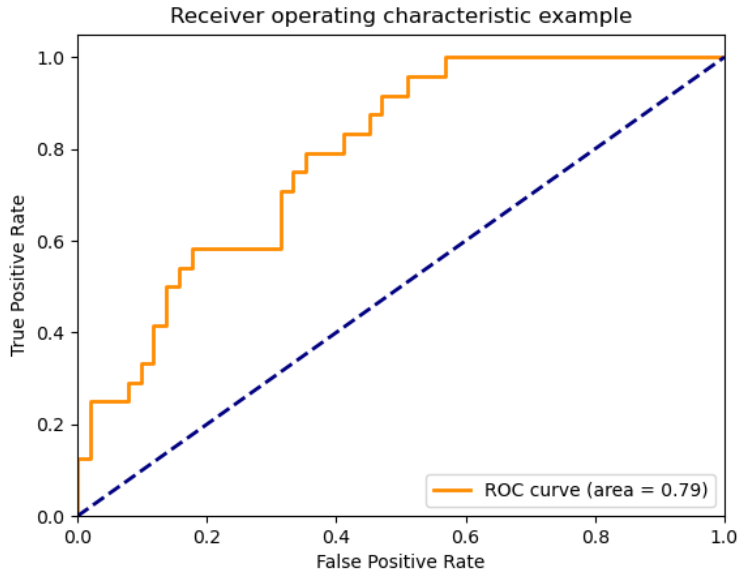
PRC plot➔ AUPRC=AP

$$Precision = \frac{TP}{TP + FP}$$



$$Recall = TPR = \frac{TP}{TP + FN}$$

ROC plot➔ AUROC



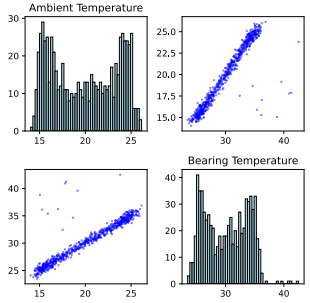
$$FPR = \frac{FP}{TN + FP}$$

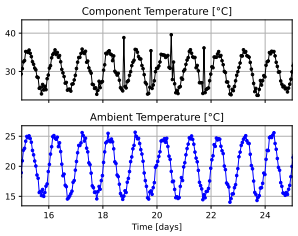
Think:

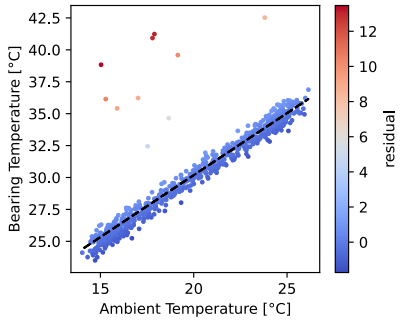
- In AD we typically have a very imbalanced data.
- Which metric is more sensitive (can compare better between similar models)?

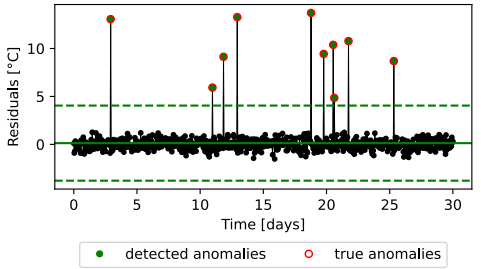
Summary AD with statistical thresholds

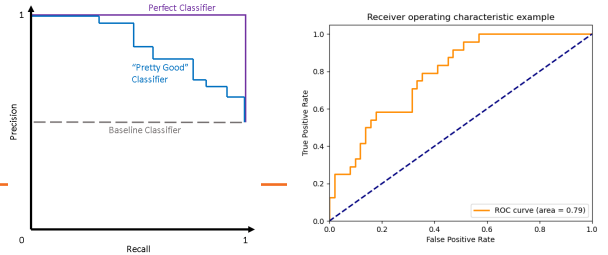
- Step 1: visualize raw variables.
- Step 2: visualize correlations.
- Step 3: model the "usual" behaviour.
- Step 4: calculate residuals and use them to derive anomaly scores.
- Step 5: select a detection threshold / decision boundary.
- Step 6: visualize the detection results as sanity check.
- Step 7: evaluate (with labels or domain knowledge).











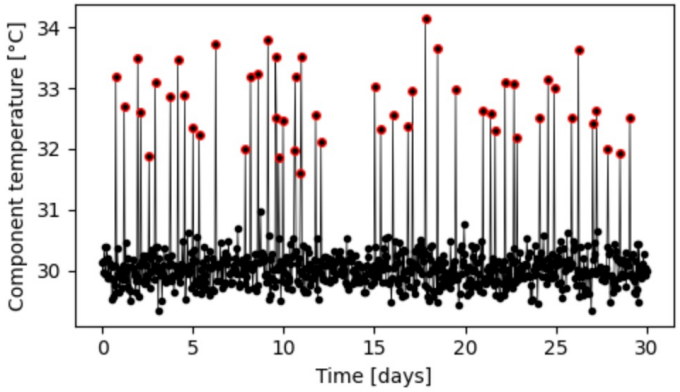
Drawbacks of the statistical approach

- We assume the data was generated by a distribution – may be wrong.
- Generally not robust to outliers: mean, std are shifted by outliers.
- More dimensions ➔ becomes harder to model.

➔Alternatives

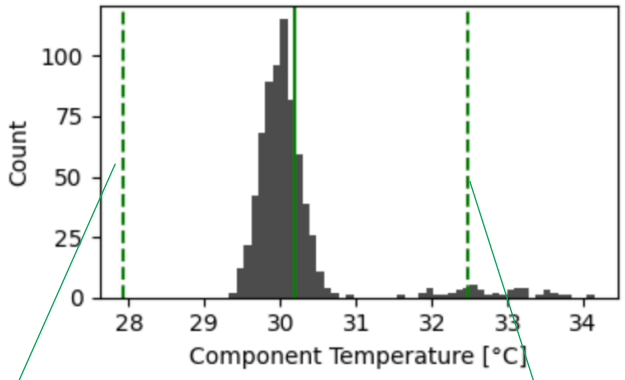
- Robust statistics: median, quantiles, robust regression.
- Non-parameteric models of the data (see later on).

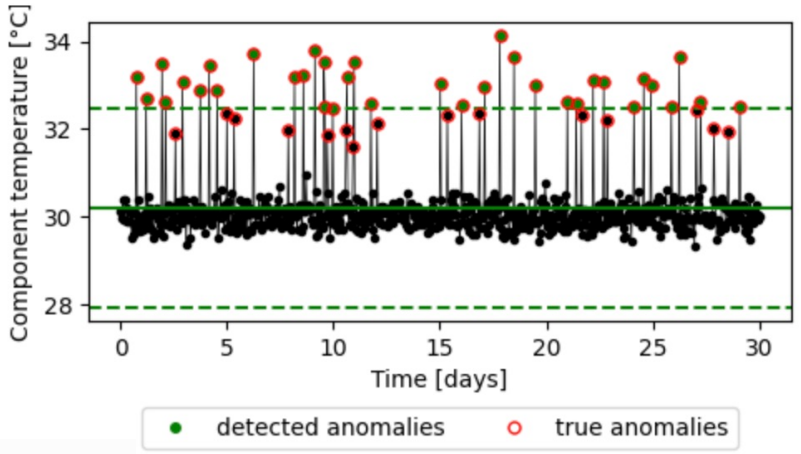
Example with more outliers



Gaussian threshold

estimate of the center $\hat{\mu}$





$T_{\text{lower}} = \mu - k\sigma$

$T_{\text{upper}} = \mu + k\sigma$

Robust thresholding

Robust estimate of the center

$$\tilde{x} = \text{median}(x_1, \ldots, x_n)$$

Median Absolute Deviation

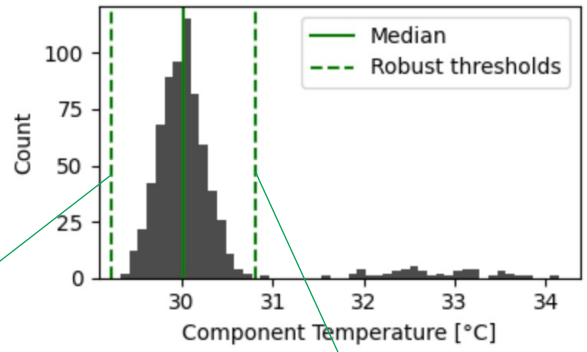
$$\text{MAD} = \text{median}\left(|x_i - \tilde{x}|\right)$$

Robust estimate of the standard deviation

$$\hat{\sigma}_{\text{robust}} = 1.4826 \cdot \text{MAD}$$

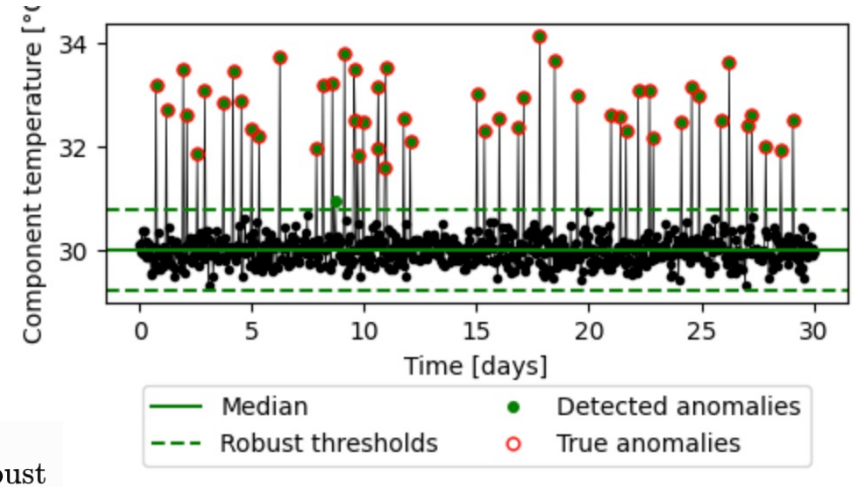
makes the estimate approximately equal to the standard deviation for Gaussian data.

Robust threshold



$$T_{\text{lower}} = \tilde{x} - k\,\hat{\sigma}_{\text{robust}}$$

$$T_{\text{upper}} = \tilde{x} + k\,\hat{\sigma}_{\text{robust}}$$



Robust regression

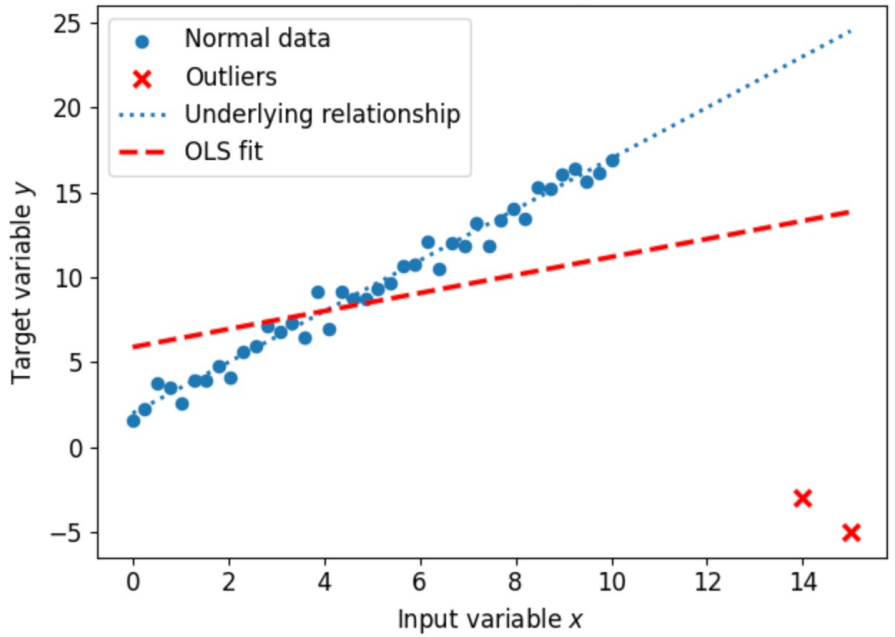
- Anomaly detection: fit a regression model despite outliers
- Ordinary Least Squares (OLS) minimizes the squared residuals

$$L = \sum_i r_i^2, \qquad r_i = y_i - \hat{y}_i.$$

- Problem: large residuals contribute quadratically

➔A few outliers can dominate the solution

➔Outliers have a disproportionately large influence on the fitted model.



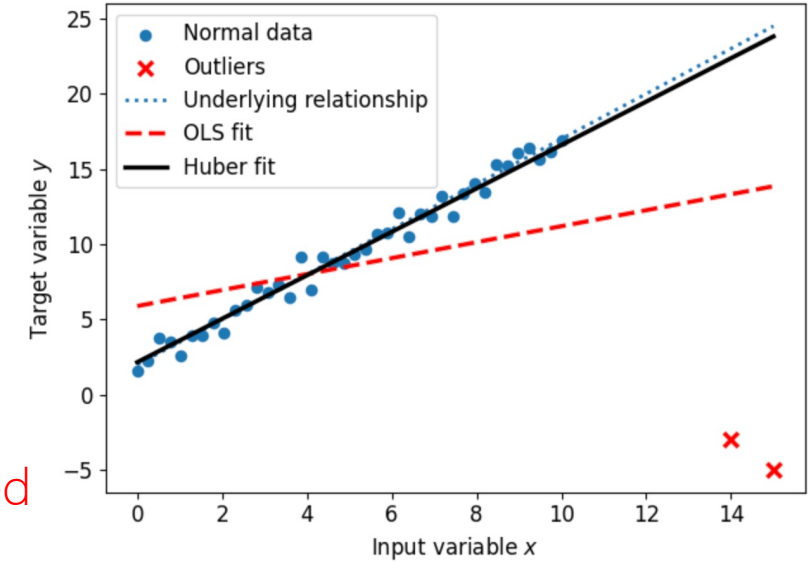
Instead of minimizing squared errors, minimize the **Huber loss**

$$L = \sum_i \ell_\delta(r_i),$$

$$\ell_\delta(r) = \begin{cases} \frac{1}{2}r^2, & |r| \le \delta, \\ \delta\left(|r| - \frac{1}{2}\delta\right), & |r| > \delta. \end{cases}$$

Small residuals → quadratic loss (like OLS)

Large residuals → linear loss (reduced influence)



➔ Outliers are not ignored but down-weighted

Instead of minimizing squared errors, minimize the **Huber loss**

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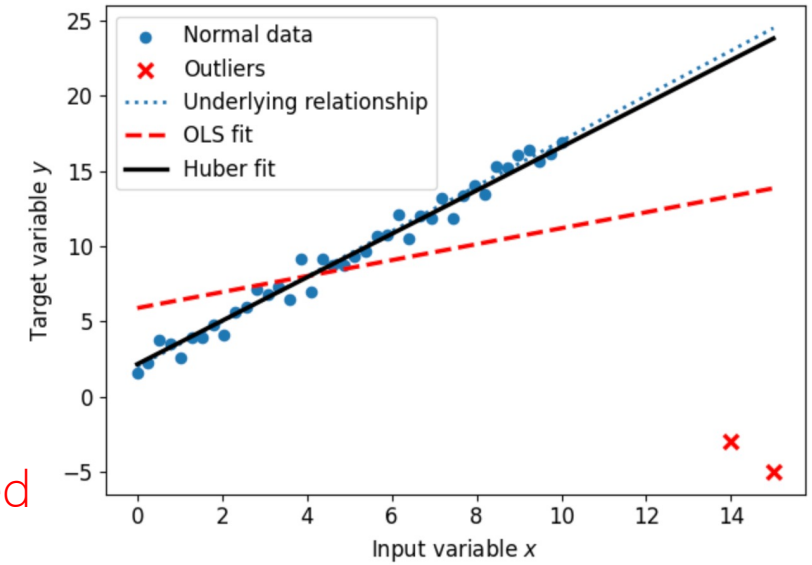
Small residuals → quadratic loss (like OLS)

Large residuals → linear loss (reduced influence)

Scikit-learn uses ϵ as a parameter $\delta = \epsilon\sigma$

Empirical std

Huber (1964) recommended $\epsilon \sim 1.35$ to retain the OLS efficiency + downweigh outliers. Use higher ϵ with more outliers.



➔ Outliers are not ignored but down-weighted

More robust regression methods

RANSAC

- Randomly select a few observations.
- Fit a regression model.
- Count how many points agree with it (the inliers).
- Repeat many times.
- Keep the model with the largest consensus set.

➔ Can tolerate even 50% outliers

Theil-Sen estimator

- Compute the slope between every pair of observations
- Take the median of these slopes.

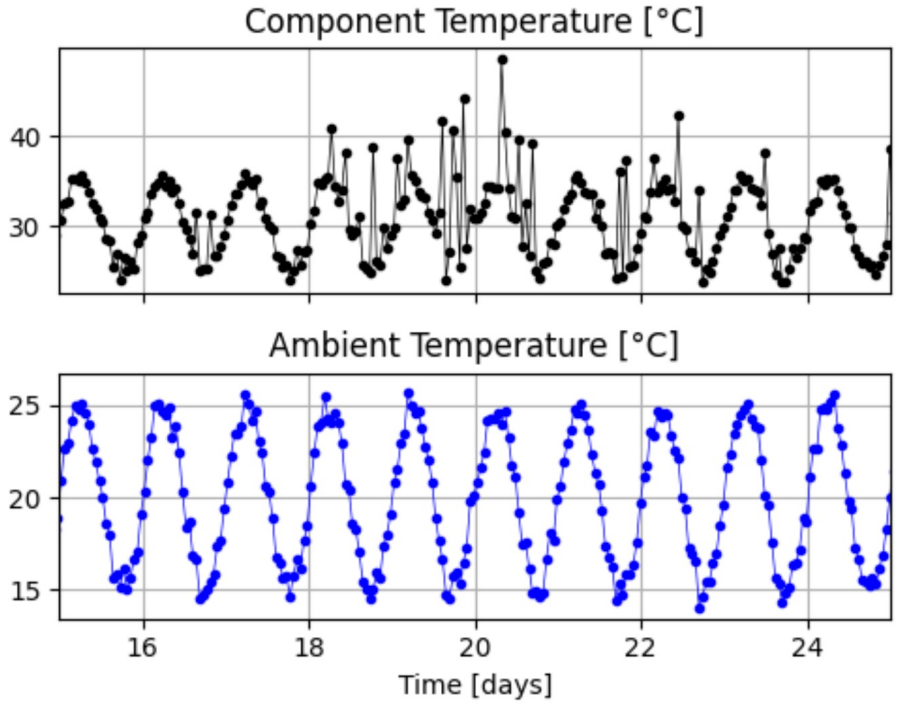
➔ Very robust but computationally expensive in high dimensions.

Exercise: try out all three robust methods on the toy problem

WPI_AnoDet_01c_Robust_regression_example

Compare linear regression to robust linear regression

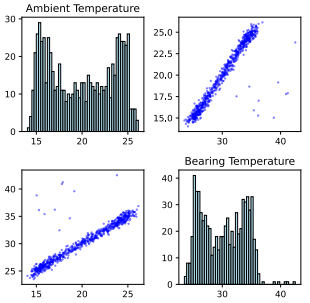
- Standard OLS regression
- Robust Huber regression
- Performance evaluation with PRC and ROC (assuming labels are given)

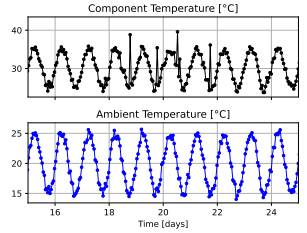


WPI_AnoDet_01d_AD_statistical_solution_with_evaluation

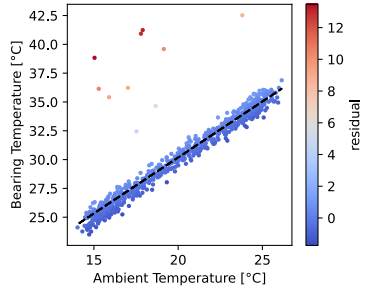
Summary AD with statistical thresholds

- Step 1: **visualize** raw variables.
- Step 2: **visualize** correlations.
- Step 3: **model** the "usual" behaviour.
  - Linear regression
  - Robust linear regression → inherently better for AD

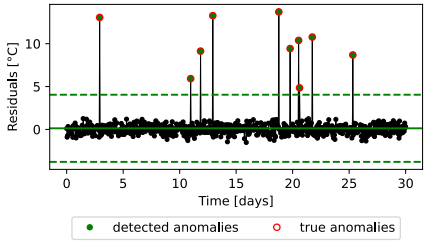




- Step 4: calculate **residuals** and use them to derive anomaly scores.



- Step 5: select a detection **threshold** / decision boundary.
  - Gaussian threshold
  - Robust threshold



- Step 6: **visualize** the detection results as sanity check.
- Step 7: **evaluate** (with labels or domain knowledge).
  - Threshold dependent: confusion matrix, precision, recall
  - Threshold independent: **ROC, PRC**

